Student Performance Analysis

Personal Data Analytics Project | Microsoft Excel

1. Project Overview

This project explored factors associated with students' final examination performance using a dataset obtained from Kaggle. The analysis examined how selected academic habits and surrounding circumstances related to students' final exam scores.

2. Project Objective

The objective was to identify patterns in student performance, investigate relationships between selected variables and final examination scores, and determine which factors were associated with differences in student performance. The findings were intended to inform discussions about academic support.

3. Dataset

The project used the *Student Performance & Study Habits Dataset*, published on Kaggle by harshadapatil31. The workbook contained 1,000 student records and 12 columns covering student characteristics, study habits, access to resources, and academic performance.

4. Tools and Methods

Microsoft Excel was used to inspect and validate the dataset, calculate descriptive statistics, create PivotTables, compare average examination scores across student groups, and develop visualizations. Histograms, scatter plots, column charts, and an interactive dashboard were used to explore patterns and communicate findings.

5. Key Findings

From my analysis, I found that students' examination performance varied across the factors I investigated. Students with internet access had an average score of 84.32, compared with 79.29 for those without access, a difference of 5.03 points. Students without part-time jobs also had a higher average score of 84.62, compared with 81.35 for students with jobs. In contrast, the differences across extracurricular activity groups were small, at just 0.46 points, while average scores across parental education levels ranged from approximately 83.29 to 85.08. I also found a positive relationship between study time and final examination scores, although study time alone did not explain all the variation in performance.

6. Conclusion

Overall, my analysis suggests that student performance is associated with a combination of academic and surrounding factors. Internet access and part-time employment showed more noticeable differences in average scores than extracurricular activities and parental education in this dataset. Study time was also positively associated with examination performance. These findings helped me see why it is important to examine several factors rather than assume that one factor alone explains students' results. However, the analysis shows associations and differences, not proof that any of these factors directly causes higher or lower examination scores.

7. Deliverables

- An Excel workbook containing the dataset analysis, PivotTables, visualizations, and interactive dashboard.

- An analysis and recommendations worksheet summarizing findings and their practical implications.
- Supporting documentation describing the project objective, analytical approach, data source, and limitations.